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(54) IMPROVED SERVOMECHANICAL CONTROL SYSTEM AND METHOD

VERBESSERTES VERFAHREN UND SYSTEM ZUR SERVOMECHANISCHEN STEUERUNG

PROCEDE ET SYSTEME DE COMMANDE DE SERVOMECHANISME AMELIORES

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US-A- 5 506 776 **US-A- 5 767 646**

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Description

[0001] The present invention relates to a method of controlling a servomechanism and servomechanism control systems, and more particularly, to a novel servomechanism control system and method featuring automatic servo loop parameter adjustment during movement, according to speed and position tracking error information.

BACKGROUND OF THE INVENTION

[0002] In the past few years, motor drive technology has switched from analog to digital control, and this change in the design approach achieves better noise immunity, and allows implementation of sophisticated drives for AC and DC motors at lower cost. Closed-loop servomechanism control systems applied to electrical motors are arranged as a feedback loop which comprise certain basic elements, including an error detector, which generates an error signal representing the difference between the command signal and the controlled output, and a controller, which amplifies the error signal. The error detection is based on use of a feedback sensor which is used to measure the state of the system, and feed the value back to the control system. For example, in the case of electrical motor control, a position feedback device mounted on the motor shaft provides position information. Another approach to motor control relates to control of the electrical current. Many other control systems can be designed using this approach.

[0003] Transient analysis of servomechanism control systems provides a definition of certain basic concepts regarding system performance, and PID (proportional-integral-derivative) control is based on these concepts. The typical issues affecting design of these systems relate to the speed of response to disturbances in a closed-loop feedback system, the amount of overshoot which can be tolerated, and the steady state error which obtains after the system has reached equilibrium.

[0004] In any servomechanism control system design, the introduction of PID control presents constraints on the achievement of good speed of response, while minimizing overshoot, and avoiding oscillations which cause system instability. Thus, achieving a well-designed servomechanism control system becomes a complicated goal when considering the need to simultaneously achieve accurate, automatic, continuous and instantaneous performance. The tightness of control is the goal, and the methods of achieving it include raising the amplitude of the response to a given error, which achieves more rapid response, but with this comes the oscillation problem.

[0005] In US-A-5 506 776 there is described a method of controlling a servomechanism according to the preamble of claim 1. A servomechanism control system according to the preamble of claim 10 is also known from US-A-5 506 776. More specifically, US-A-5 506 776 discloses a method generating an error signal in a feedback loop having a variable gain to actuate the driven element to reduce the error signal, wherein the gain can have only two values, 0 and a predefined value, depending whether the position error absolute value is greater than a predefined limit or not.

[0006] Therefore, it would be desirable to provide a method which improves the tightness of control for servomechanism, without unduly complicating the apparatus and its operation.

[0007] Accordingly, it is a principal object of the present invention to overcome prior art problems associated with PID control of servomechanisms and the speed of response, and provide an improved method for controlling a servomechanism.

[0008] To achieve this, the method of the invention is characterized by the features claimed in the characterizing part of claim 1 and the invention provides a control system according to the characterizing part of claim 10.

[0009] Basically, in accordance with the present invention there is provided an improved method of controlling a servomechanism, characterized in that the variable gain is provided in accordance with a growing function related to both the absolute value of the error signal and the absolute value of the velocity of the command signal, wherein the variable gain function increases with the absolute value of the error signal and the absolute value of the velocity of the command signal, and has a finite maximum value and positive minimum value.

[0010] The inventive method is applicable to nearly all types of servomechanism control systems. In general application, it is useful to control the state of a system, using an actuator, which provides a means of changing the state of the system, and a feedback sensor, which provides a means to measure the state of the system, to provide a feedback value to the control system. Examples of these types of systems include servomechanism control of an electrical motor with a position feedback device mounted on the shaft; a machine activated by a hydraulic piston with a position feedback device; control of the electrical current in an electrical motor; control of the water level in a boiler; control of aeronautical equipment such as aircraft wing flap position systems; valve control in chemical process plants, etc.

[0011] The inventive method is applicable in all of these types of systems to reduce the tracking error, i.e., the difference between the desired state and the actual state of the controlled variable. It also increases the speed of response, reducing the time necessary for the system to reach a new state value within a given precision. Further, it increases control tightness, i.e., it reduces the tracking error induced by external perturbations.

[0012] A feature of the inventive method is its ability to improve the performance of servomechanism control systems which utilize a digital feedback sensor. Unlike with prior art systems, which have limited gain when using a digital

feedback sensor, the inventive method enables the gain to be varied, by raising it only when necessary, such as when a perturbation occurs, or when the the speed is increased.

[0013] For example, in prior art systems, such as a position control system using an electrical motor with an optical encoder, the discrete nature of the encoder reduces the maximum loop gain of the system at very low speeds. At low speeds, the speed sensing based on the encoder is not accurate, and raising the loop gain produces strong vibrations around the equilibrium position. The inventive method allows for a variable gain, thus enabling tighter control.

[0014] Other features and advantages of the invention will become apparent from the following drawings and the description.

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] For a better understanding of the invention with regard to the embodiments thereof, reference is made to the accompanying drawings, in which like numerals designate corresponding elements or sections throughout, and in which:

Fig. 1 is a schematic block diagram of a simplified servomechanism control system; and

Figs. 2a-b illustrate a flowchart of a method of controlling the system of Fig. 1 in accordance with the method of the present invention.

DETAILED DESCRIPTION OF A PREFERRED EMBODIMENT

[0016] Referring now to Fig. 1, there is shown a schematic block diagram of a simplified servomechanism control system 10, for example, in an application providing position control using an electric motor as the system actuator. Control system 10 comprises a closed-loop feedback system having a command position generator 12, an error detector 14, a controller 16, an actuator 18, and a position feedback device 20. The controller 16 is shown with a control equation written in the form of a PID control loop, expressed as a Laplace transform. The control variables shown in Fig. 1 are defined as follows:

C Command position, which is the position the system is instructed to reach.

V_c Command speed, which is the speed of the position command.

E Actual position, which is the present position of the system, such as the shaft angle of an electrical motor.

ε Position error = (Command Position) - (Actual Position)

I Feedback actuator value (e.g., current of an electrical motor)

[0017] In most conventional systems, control equations can be written in the PID control loop form as follows:

$$I = K_i * \int \epsilon * dt + K_p * \epsilon + K_d * d\epsilon/dt \quad (1)$$

Many variations, additions and more elaborate algorithms exist, with most having the same basic parameters defined (the PID coefficients):

K_i Integral Feedback Coefficient

K_p Proportional Feedback Coefficient

K_d Differential Feedback Coefficient

In many cases the equations are written in a different format, but by mathematical transformation they can be written in the same way, sometimes with additional terms. A common technique of expressing the equation uses the Laplace Transform, in which F(s) represents the transfer function of the physical system, as shown in Fig. 1.

[0018] In accordance with the principles of the present invention, the improved method of servomechanism control utilizes the same form of the equation (1), but the parameters of the equation are adjusted in a specific way, as described further herein.

[0019] In order to describe the algorithm utilized in the improved method of control per the invention, the following portion of the description provides definitions for the control variables, their relationships and effect upon the improved control method.

1) K_g is defined as a variable used to change the gain of the control loop. This variable affects the gain of the PID equation (1). In the inventive method, this variable is a combined function of position error and of command speed.

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At command speed zero and position error zero, $K_g = 1$, i.e. the PID coefficients are not changed. During movement, the command speed and position error values increase. K_g will then be changed to an increased value which will cause the system to respond strongly in order to reduce position error.

2) t_KgPar is defined as a coefficient setting the rate of change of K_g with command speed. This coefficient is given a fixed value for a given system, and it defines the relationship between K_g and the position error.

3) nKd is defined as the coefficient setting the rate of change of K_g with the command speed. This coefficient is given a fixed value for a given system, and it defines the relationship between K_g and the command speed.

4) $MinSpeed$ is defined as a speed threshold parameter. This coefficient is given a fixed value for a given system, and it defines the relationship of K_g and the command speed. When the command speed is well under the given fixed value, K_g is independent of the command speed. When the command speed exceeds that value, K_g increases progressively with the command speed.

5) $MaxKg$ is defined as an upper limit for K_g . Variation of K_g with the command speed and position error is limited to that maximum value.

6) $KgFilter$ is defined as a parameter for filtering K_g , when K_g is decreasing. In order to avoid discontinuity of the feedback response, a low pass time filter is applied on K_g when the K_g value is decreasing. No filter is applied when K_g is increasing.

[0020] With the above definitions, the following equation can be presented in accordance with the inventive control method, representing the amount of feedback applied to the servomechanism control system (herein: Integral feedback)

$$I = \text{Integral} + K_p.K_g^2.\epsilon + K_d.K_g.d\epsilon/dt \quad (2)$$

where Integral is defined as follows:

$$\text{Integral}(t+dt) = \text{Integral}(t) + dt \cdot K_{ii}.K_g.K_g.K_g.\epsilon + dt.K_{ip}.K_g.K_g.d\epsilon/dt \quad (3)$$

K_{ii} and K_{ip} are defined as follows: (where $|\epsilon|$ represents the absolute value of ϵ)

If ($|\epsilon| > 0$) and ($d|\epsilon|/dt > 0$), then [$K_{ii} = K_{is}$, $K_{ip} = K_{iv}$]

Otherwise [$K_{ii} = K_i$, $K_{ip} = 0$].

$$K_g \text{ is calculated by the following formula: Initial value of } K_g, K_g = 1 \quad (4)$$

$$TKg = f(|\epsilon|, |Vcl|) \quad (5)$$

where f is a growing function of both position error $|\epsilon|$ and command speed $|Vcl|$, with a finite maximum value, and a positive minimum value of 1. Many mathematical functions or tables of values conforming to these rules may be chosen for f , and for the preferred embodiment the following function was chosen:

$$TKg1 = \{nKd * abs|Vcl| + MinSpeed\} / \{|Vcl| + MinSpeed\} + t_KgPar * |\epsilon| \quad (6)$$

$$\text{If } (TKg > MaxKg), \text{ then } TKg = MaxKg \quad (7)$$

$$\text{If } (TKg > Kg) \text{ then } Kg = TKg \quad (8)$$

$$\text{Otherwise, } Kg(t+dt) = Kg(t) + KgFilter*dt*(TKg - Kg(t)) \quad (9)$$

[0021] In accordance with the inventive control method, the above control equations (1)-(9) provide the following functions:

(1) Function of K_g :

[0022] The feedback parameters are increased by K_g . At equilibrium, i.e., when the position error is small, and when the speed is low, smooth operation is generally required, and in such cases, it is desirable to limit the feedback gain, so as to avoid vibrations in the control system. Another reason for limiting the feedback gain at low speeds is that most servo control systems have non-linear behavior at steady state, due to friction, the digital nature of feedback devices, backlash of gears and other physical limitations. In many cases, these physical limitations limit the maximum feedback gain of the control system at steady state. However, during movement, a much higher feedback gain can be allowed.

[0023] In accordance with the principles of the inventive method, K_g makes it possible to change the gain during movement, and the specific combination of command speed and position error variables in the K_g formula (equations 4-9 above) provides a practical way of changing the gain without introducing oscillations in the control system.

[0024] Also, the value of K_g is raised to a different mathematical exponent for each feedback parameter, and this maintains the control loop stability for all K_g values in the range $< \text{Max}K_g$.

(2) Function of K_i , K_v :

[0025] In conventional systems, integral feedback is proportional to the position error integral. If the integral feedback gain is increased, then the problem of overshoot is present and system instability is obtained. In the scope of the present invention, it was observed that when position error is decreasing, the integral has reached a sufficient value to compensate the perturbations that caused the position error. However, in conventional systems, the integral term continues to grow, and when the position error finally returns to zero, the integral term has too large a value, and is the source of overshoot. This is the main reason that limits are placed on the integral feedback gain.

[0026] In accordance with the present invention, the integral feedback term grows strongly when the position error is growing, but grows slowly (or stops when $K_i=0$), when the position error is decreasing. This eliminates the causes of the overshoot, and thus the integral feedback gain can have a much larger value, as large as 10 times more.

[0027] Similarly, the inventive method applies this approach to the integral of the position error derivative (K_v). Since the integral of the position error derivative is related to the position error, K_v increases the so-called proportional feedback gain K_p without creating oscillations.

[0028] Referring now to Figs. 2a-b, there is illustrated a flowchart of a method of controlling the system of Fig. 1 in accordance with the method of the present invention for example, in an application providing position control using an electric motor as the system actuator. In the preferred embodiment, the method is incorporated in a software control routine, which is typically run on a control hardware computer system, which may be implemented on an NEC V853 microprocessor chip, or other, in accordance with skill of the art software programming techniques.

[0029] The control routine described in Figs. 2a-b is run on the control hardware, and in block 25, the control routine is called repetitively, typically by an interrupt or delay in order to call the routine at regular intervals. In block 28, the time interval between the present call and the last call of the control routine is calculated. Typically, if regular intervals are used, the variable dt can be set to a constant value.

[0030] In block 30, in order to calculate speed, the previous value of the position is stored in memory. In block 32, the present command position and system position are measured, and the difference is calculated, resulting in position error ϵ . These values of command position and system position are read via interface devices, provided in the specific application, such as digital encoders.

[0031] In block 34, a target value for the gain factor (K_g) is calculated, using the formulas in equations (5) and (6). This formula increases the gain progressively with position error and command speed. In block 36, the target value for the gain is tested, and if the target value for gain is lower than the present one, a filter is applied to K_g in block 38. This leaves a higher gain for some time after a perturbation, thus giving the system more time to stabilize before lowering the gain back to its initial value.

[0032] If the block 36 test of the target value for gain shows that it is higher than the present one, in block 40 the gain immediately jumps to a high value in order to respond quickly and strongly to a detected perturbation. In block 42, the proportional and differential feedback is calculated in accordance with the formula shown. The proportional feedback is multiplied by the gain K_g at exponent 2, and the differential feedback is multiplied by the gain K_g .

[0033] In block 44, the position error is tested, to determine whether it is increasing or decreasing, and the control routine selects the values for the integral feedback parameters accordingly. If the position error is increasing, then in block 46, the feedback gain is increased, and a high value of the Integral feedback (equation (2)) is chosen.

[0034] If, however, the position error is decreasing, then this means that the Integral feedback has finally reached a value that will bring the system toward the command position. If the Integral feedback were to continue to grow, then when the system finally reaches the command position, the value of the Integral feedback would be too large, and an overshoot would be created. Thus, in block 48, a low value is selected for K_{ji} and a zero value is selected for K_{ip} .

[0035] In block 50, the Integral feedback is calculated per equation (2), and in this step, K_{ii} is multiplied by K_g at

exponent 3, and Kip is multiplied by Kg at exponent 2.

[0036] In block 52, the total calculated feedback is output to the system. In block 54, the end of the control routine is reached, and the routine is called again by the system software at the next time interval dt.

[0037] In alternative embodiment, the closed-loop feedback system may be implemented in an analog signal processor, in accordance with standard design techniques.

[0038] Having described the inventive method with regard to certain specific steps, it is to be understood that the description is not meant as a limitation, since further modifications may now suggest themselves to those skilled in the art, and it is intended to cover such modifications as fall within the scope of the appended claims.

Claims

1. A method of controlling a servomechanism, said method comprising the step of:

generating (12) a command signal (C) representing a desired physical quantity,
actuating (18) a driven element to output said physical quantity (E) in accordance with said command signal,
measuring (14) said output physical quantity (E) and comparing it with said command signal,
generating an error signal (ϵ) representing the difference between said command signal (C) and said measured output (E), and
providing (16, 20) said error signal (ϵ) in a feedback loop (F(s), I) having a variable gain (Kg) to actuate said driven element in a fashion so as to reduce said error signal (ϵ) to a zero value,

characterized in that said variable gain (Kg) is provided in accordance with a growing function (TKg) related to both the absolute value of said error signal ($|\epsilon|$) and the absolute value of the velocity ($|V_c|$) of said command signal,

wherein said variable gain function (TKg) increases with said absolute value of said error signal ($|\epsilon|$) and said absolute value of the velocity ($|V_c|$) of said command signal, and has a finite maximum value and positive minimum value.

2. The method of claim 1, **characterized in that** said feedback loop (I) comprises elements representing differential feedback, proportional feedback and integral feedback.

3. The method of claim 2, **characterized in that** said variable gain (Kg) is applied to said differential feedback element with an exponent of one in said feedback loop (I).

4. The method of claim 2, **characterized in that** said variable gain (Kg) is applied to said proportional feedback element with an exponent of two in said feedback loop (I).

5. The method of claim 2, **characterized in that** said variable gain (Kg) is applied to said integral feedback element with an exponent of three in said feedback loop (I).

6. The method of claim 1, **characterized in that** said method is applied to a position control system (10) in which said driven element is provided as a motor and said variable gain function (TKg) is provided in accordance with the following equation:

$$Tkg1 = \{nKd \cdot \text{abs}|V_c| + \text{MinSpeed}\} / \{|V_c| + \text{MinSpeed}\} + t_KgPar \cdot |\epsilon|$$

where , If (Tkg > MaxKg), then Tkg=MaxKg
and where:

Vc=Command speed, which is the speed of the position command,

MinSpeed = a speed threshold parameter,

MaxKg = an upper limit for Kg,

T_KgPar = a coefficient setting the rate of change of Kg with command speed,

ϵ = Position error

7. The method of claim 1, **characterized in** further comprising the step of filtering (36) said variable gain function

(TKg) in asymmetric fashion by applying a low pass time filter (38) to said variable gain (Kg) when its value is decreasing, not when its value is increasing.

8. The method of claim 6, **characterized in that** said error signal (ϵ) comprises a position error and said feedback loop (Tkg1) comprises an integral feedback element which is adjusted in accordance with a derivative of the absolute value of said position error signal (ϵ), such that said integral feedback element has a relatively high value when said position error signal (ϵ) is increasing, and a relatively low value when said position error signal (ϵ) is decreasing.

9. The method of claim 6, **characterized in that** said error signal (ϵ) comprises a position error and said feedback loop (Tkg1) comprises a velocity error integral feedback element which is adjusted in accordance with a derivative of the absolute value of said position error signal ($|\epsilon|$), such that said integral feedback element has a relatively high value when said position error signal (ϵ) is increasing, and a relatively low value when said position error signal (ϵ) is decreasing.

10. A servomechanism control system (10) comprising:

means (12) for generating a command signal (C) representing a desired physical quantity,

means (18) for actuating a driven element to output said physical quantity (E) in accordance with said command signal,

means (14) for measuring said output physical quantity (E) and comparing it with said command signal (C),

means for generating an error signal (ϵ) representing the difference between said command signal (C) and said measured output (E), and

means (16, 20) for providing said error signal (ϵ) in a feedback loop (F(s), I) having a variable gain (Kg) to actuate said driven element in a fashion so as to reduce said error signal (ϵ) to a zero value,

characterized in that said variable gain (Kg) is provided in accordance with a growing function (TKg) related to both the absolute value of said error signal ($|\epsilon|$) and the absolute value of the velocity ($IV_c I$) of said command signal,

wherein said variable gain function (TKg) increases with said absolute value of said error signal ($|\epsilon|$) and said absolute value of the velocity of said command signal ($IV_c I$), and has a finite maximum value and positive minimum value.

11. The control system of claim 10, **characterized in that** said means (16, 20) for providing said error signal in a feedback loop comprises a processing unit implemented in a microcontroller programmed with control software.

12. The control system of claim 10, **characterized in that** said means (16, 20) for providing said error signal in a feedback loop comprises a processing unit implemented in an analog signal processor.

Patentansprüche

1. Ein Verfahren zur Steuerung eines Servomechanismus, wobei besagtes Verfahren den Schritt umfasst des:

Generierens (12) eines Befehlssignals (C), das eine erwünschte physikalische Qualität darstellt,

Auslösens (18) eines angetriebenen Elements, um besagte physikalische Quantität (E) in Übereinstimmung mit besagtem Befehlssignal abzugeben,

Messens (14) besagter abgegebener physikalischer Quantität (E) und deren Vergleichens mit besagtem Befehlssignal,

Generierens eines Fehlersignals (ϵ), das die Differenz zwischen besagtem Befehlssignal (C) und besagtem gemessenem Output (E) darstellt, und des

Vorsehens (16, 20) besagten Fehlersignals (ϵ) in einer Feedbackschleife ((F(s), I) mit einem variablen Stellfaktor (Kg), um besagtes angetriebenes Element auf eine Weise auszulösen, dass besagtes Fehlersignal (ϵ) auf einen Nullwert reduziert wird,

dadurch gekennzeichnet, dass besagter variabler Stellfaktor (Kg) in Übereinstimmung mit einer wachsenden Funktion (TKg) vorgesehen wird, die sowohl auf den absoluten Wert besagten Fehlersignals ($|\epsilon|$) als auch den absoluten Wert der Geschwindigkeit ($IV_c I$) besagten Befehlssignals bezogen ist,

wobei besagte variable Stellfaktorfunktion (TKg) mit besagtem absoluten Wert besagten Fehlersignals ($|e|$) und besagtem absoluten Wert der Geschwindigkeit ($|V_c|$) besagten Befehlssignals ansteigt und einen finiten Maximalwert und positiven Minimalwert besitzt.

2. Das Verfahren gemäß Anspruch 1, **dadurch gekennzeichnet, dass** besagte Feedbackschleife (I) Elemente umfasst, die differentielles Feedback, proportionales Feedback und integrales Feedback darstellen.
3. Das Verfahren gemäß Anspruch 2, **dadurch gekennzeichnet, dass** besagter variabler Stellfaktor (Kg) mit einem Exponenten von eins in besagter Feedbackschleife (I) auf besagtes differentielles Feedbackelement angewendet wird.
4. Das Verfahren gemäß Anspruch 2, **dadurch gekennzeichnet, dass** besagter variabler Stellfaktor (Kg) mit einem Exponenten von zwei in besagter Feedbackschleife (I) auf besagtes proportionales Feedbackelement angewendet wird.
5. Das Verfahren gemäß Anspruch 2, **dadurch gekennzeichnet, dass** besagter variabler Stellfaktor (Kg) mit einem Exponenten von drei in besagter Feedbackschleife (I) auf besagtes integrales Feedbackelement angewendet wird.
6. Das Verfahren gemäß Anspruch 1, **dadurch gekennzeichnet, dass** besagtes Verfahren auf ein Positionskontrollsystem (10) angewendet wird, wobei besagtes angetriebenes Element als Motor vorgesehen ist und besagte variable Stellfaktorfunktion (TKg) in Übereinstimmung mit der folgenden Gleichung vorgesehen ist:

$$TKg1 = \{nKd \cdot \text{abs}|V_c| + \text{MinGeschw}\} / \{|V_c| + \text{MinGeschw}\} + t_KgPar \cdot |e|,$$

wobei, wenn $(TKg > \text{MaxKg})$, dann $TKg = \text{MaxKg}$,
und wobei:

V_c = Befehlsgeschwindigkeit, was die Geschwindigkeit des Positionsfehlers ist,
 MinGeschw = ein Geschwindigkeits-Schwellenparameter,
 MaxKg = eine Obergrenze für Kg,
 T_KgPar = ein Koeffizient, der die Veränderungsrate von Kg mit der Befehlsgeschwindigkeit bestimmt,
 e = Positionsfehler.

7. Das Verfahren gemäß Anspruch 1, **dadurch gekennzeichnet, dass** es weiterhin den Schritt des Filterns (36) besagter variabler Stellfaktorfunktion (TKg) auf asymmetrische Weise durch Anwendung eines Tiefpasszeitfilters (38) auf besagten variablen Stellfaktor (Kg), wenn dessen Wert abnimmt, nicht wenn dessen Wert zunimmt, umfasst.
8. Das Verfahren gemäß Anspruch 6, **dadurch gekennzeichnet, dass** besagtes Fehlersignal (e) einen Positionsfehler umfasst und besagte Feedbackschleife (TKg1) ein integrales Feedbackelement umfasst, das in Übereinstimmung mit einer Ableitung des absoluten Werts besagten Positionsfehlersignals (e) eingestellt ist, sodass besagtes integrales Feedbackelement einen relativ hohen Wert hat, wenn besagtes Positionsfehlersignal (e) ansteigt, und einen relativ niedrigen Wert, wenn besagtes Positionsfehlersignal (e) abnimmt.
9. Das Verfahren gemäß Anspruch 6, **dadurch gekennzeichnet, dass** besagtes Fehlersignal (e) einen Positionsfehler umfasst und besagte Feedbackschleife (TKg1) ein integrales Geschwindigkeitsfehler-Feedbackelement umfasst, das in Übereinstimmung mit einer Ableitung des absoluten Werts besagten Positionsfehlersignals ($|e|$) eingestellt ist, sodass besagtes integrales Feedbackelement einen relativ hohen Wert hat, wenn besagtes Positionsfehlersignal (e) ansteigt, und einen relativ niedrigen Wert, wenn besagtes Positionsfehlersignal (e) abnimmt.
10. Ein Servomechanismus-Steuersystem (10), umfassend:
Mittel (12) zum Generieren eines Befehlssignals (C), das eine erwünschte physikalische Quantität darstellt,
Mittel (18) zum Auslösen eines angetriebenen Elements, um besagte physikalische Quantität (E) in Übereinstimmung mit besagtem Befehlssignal abzugeben,

Mittel (14) zum Messen besagter abgegebener physikalischer Quantität (E) und deren Vergleichen mit besagtem Befehlssignal (C),

Mittel zum Generieren eines Fehlersignals (ϵ), das die Differenz zwischen besagtem Befehlssignal (C) und besagtem gemessenem Output (E) darstellt, und

Mittel (16, 20) zum Bereitstellen besagten Fehlersignals (ϵ) in einer Feedbackschleife ($F(s)$, I) mit einem variablen Stellfaktor (Kg), um besagtes angetriebenes Element auf eine Weise auszulösen, dass besagtes Fehlersignal (ϵ) auf einen Nullwert reduziert wird,

dadurch gekennzeichnet, dass besagter variabler Stellfaktor (Kg) in Übereinstimmung mit einer wachsenden Funktion (TKg) vorgesehen wird, die sowohl auf den absoluten Wert besagten Fehlersignals ($|\epsilon|$) als auch den absoluten Wert der Geschwindigkeit (IV_c) besagten Befehlssignals bezogen ist, wobei besagte variable Stellfaktorfunktion (TKg) mit besagtem absoluten Wert besagten Fehlersignals ($|\epsilon|$) und besagtem absoluten Wert der Geschwindigkeit (IV_c) besagten Befehlssignals ansteigt und einen finiten Maximalwert und positiven Minimalwert besitzt.

11. Das Steuersystem von Anspruch 10, **dadurch gekennzeichnet, dass** besagtes Mittel (16, 20) zur Bereitstellung besagten Fehlersignals in einer Feedbackschleife eine in einem mit Steuersoftware programmiertem Mikrosteu-
ergerät implementierte Datenverarbeitungseinheit umfasst.

12. Das Steuersystem von Anspruch 10, **dadurch gekennzeichnet, dass** besagtes Mittel (16, 20) zur Bereitstellung besagten Fehlersignals in einer Feedbackschleife eine in einem analogen Signalverarbeitungsgerät implemen-
tierte Datenverarbeitungseinheit umfasst.

Revendications

1. Procédé de commande d'un servomécanisme, ledit procédé comprenant les étapes consistant à :

générer (12) un signal de commande (C) représentant une quantité physique désirée ;
actionner (18) un élément entraîné dans le but de transmettre ladite quantité physique (E) conformément audit
signal de commande ;
mesurer (14) ladite quantité physique transmise (E) et la comparer audit signal de commande ;
générer un signal d'erreur (ϵ) représentant la différence entre ledit signal de commande (C) et ladite sortie
mesurée (E) ; et
procurer (16, 20) ledit signal d'erreur (ϵ) dans une boucle d'asservissement ($F(s)$, I) possédant un gain variable
(Kg) afin d'actionner ledit élément entraîné de façon à réduire ledit signal d'erreur (ϵ) jusqu'à la valeur zéro,

caractérisé en ce que ledit gain variable (Kg) est fourni conformément à une fonction croissante (TKg) apparentée
à la fois à la valeur absolue dudit signal d'erreur ($|\epsilon|$) et à la valeur absolue de la vitesse (IV_c) dudit signal de
commande,
ladite fonction à gain variable (TKg) augmentant de manière proportionnelle à ladite valeur absolue dudit signal
d'erreur ($|\epsilon|$) et à ladite valeur absolue de la vitesse (IV_c) dudit signal de commande, et possédant une valeur
maximale finie et une valeur minimale positive.

2. Procédé selon la revendication 1, **caractérisé en ce que** ladite boucle d'asservissement (I) comprend des élé-
ments représentant une rétroaction différentielle, une rétroaction proportionnelle et une rétroaction intégrale.

3. Procédé selon la revendication 2, **caractérisé en ce que** ledit gain variable (Kg) est appliqué audit élément de
rétroaction différentielle avec un exposant de un dans ladite boucle de rétroaction (I).

4. Procédé selon la revendication 2, **caractérisé en ce que** ledit gain variable (Kg) est appliqué audit élément de
rétroaction proportionnelle avec un exposant de deux dans ladite boucle de rétroaction (I).

5. Procédé selon la revendication 2, **caractérisé en ce que** ledit gain variable (Kg) est appliqué audit élément de
rétroaction intégrale avec un exposant de trois dans ladite boucle de rétroaction (I).

6. Procédé selon la revendication 1, **caractérisé en ce que** ledit procédé est appliqué à un système de commande
de position (10) dans lequel ledit élément entraîné est prévu sous la forme d'un moteur et ladite fonction à gain

variable (TKg) est prévue conformément à l'équation ci-après :

$$Tkg1 = \{nKd \cdot \text{abs} |Vcl| + \text{MinSpeed}\} / \{|Vcl| + \text{MinSpeed}\} + t_KgPar \cdot |e|$$

dans laquelle,
lorsque (TKg > MaxKg), Tkg = MaxKg
et dans laquelle

Vc = la vitesse de commande qui représente la vitesse de la commande de position ;
MinSpeed = un paramètre de seuil de vitesse ;
MaxKg = une limite supérieure pour Kg ;
T_KgPar = un coefficient qui règle le taux de changement de Kg en fonction de la vitesse de commande ;
ε = une erreur de position.

7. Procédé selon la revendication 1, **caractérisé en ce qu'**elle comprend en outre l'étape consistant à filtrer (36) ladite fonction à gain variable (TKg) de manière asymétrique en appliquant un filtre temporel passe-bas (38) audit gain variable (Kg) lorsque sa valeur diminue, et non lorsque sa valeur augmente.

8. Procédé selon la revendication 6, **caractérisé en ce que** ledit signal d'erreur (ε) comprend une erreur de position et ladite boucle d'asservissement (Tkg1) comprend un élément de rétroaction intégrale qui est réglé conformément à une dérivée de la valeur absolue dudit signal d'erreur de position (e), de telle sorte que ledit élément de rétroaction intégrale possède une valeur relativement élevée lorsque ledit signal d'erreur de position (ε) augmente, et une valeur relativement faible lorsque ledit signal d'erreur de position (ε) diminue.

9. Procédé selon la revendication 6, **caractérisé en ce que** ledit signal d'erreur (ε) comprend une erreur de position et ladite boucle d'asservissement (Tkg1) comprend un élément de rétroaction intégrale d'erreur de vitesse qui est réglé conformément à une dérivée de la valeur absolue dudit signal d'erreur de position { |e| }, de telle sorte que ledit élément de rétroaction intégrale possède une valeur relativement élevée lorsque ledit signal d'erreur de position (ε) augmente, et une valeur relativement faible lorsque ledit signal d'erreur de position (ε) diminue.

10. Système de commande de servomécanisme (10) comprenant :

des moyens (12) pour générer un signal de commande (C) représentant une quantité physique désirée ;
des moyens (18) pour actionner un élément entraîné dans le but de transmettre ladite quantité physique (E) conformément audit signal de commande ;
des moyens (14) pour mesurer ladite quantité physique transmise (E) et pour la comparer audit signal de commande ;
des moyens générer un signal d'erreur (ε) représentant la différence entre ledit signal de commande (C) et ladite sortie mesurée (E) ; et
des moyens (16, 20) pour procurer ledit signal d'erreur (ε) dans une boucle d'asservissement (F(s), I) possédant un gain variable (Kg) afin d'actionner ledit élément entraîné de façon à réduire ledit signal d'erreur (ε) jusqu'à la valeur zéro,

caractérisé en ce que ledit gain variable (Kg) est fourni conformément à une fonction croissante (TKg) apparentée à la fois à la valeur absolue dudit signal d'erreur (|e|) et à la valeur absolue de la vitesse (|Vcl|) dudit signal de commande, ladite fonction à gain variable (TKg) augmentant de manière proportionnelle à ladite valeur absolue dudit signal d'erreur (|e|) et à ladite valeur absolue de la vitesse (|Vcl|) dudit signal de commande, et possédant une valeur maximale finie et une valeur minimale positive.

11. Système de commande selon la revendication 10, **caractérisé en ce que** lesdits moyens (16, 20) pour procurer ledit signal d'erreur dans une boucle d'asservissement comprennent une unité de traitement implémentée dans un microcontrôleur programmé avec un logiciel de commande.

12. Système de commande selon la revendication 10, **caractérisé en ce que** lesdits moyens (16, 20) pour procurer ledit signal d'erreur dans une boucle d'asservissement comprennent une unité de traitement implémentée dans un processeur de signaux analogiques.

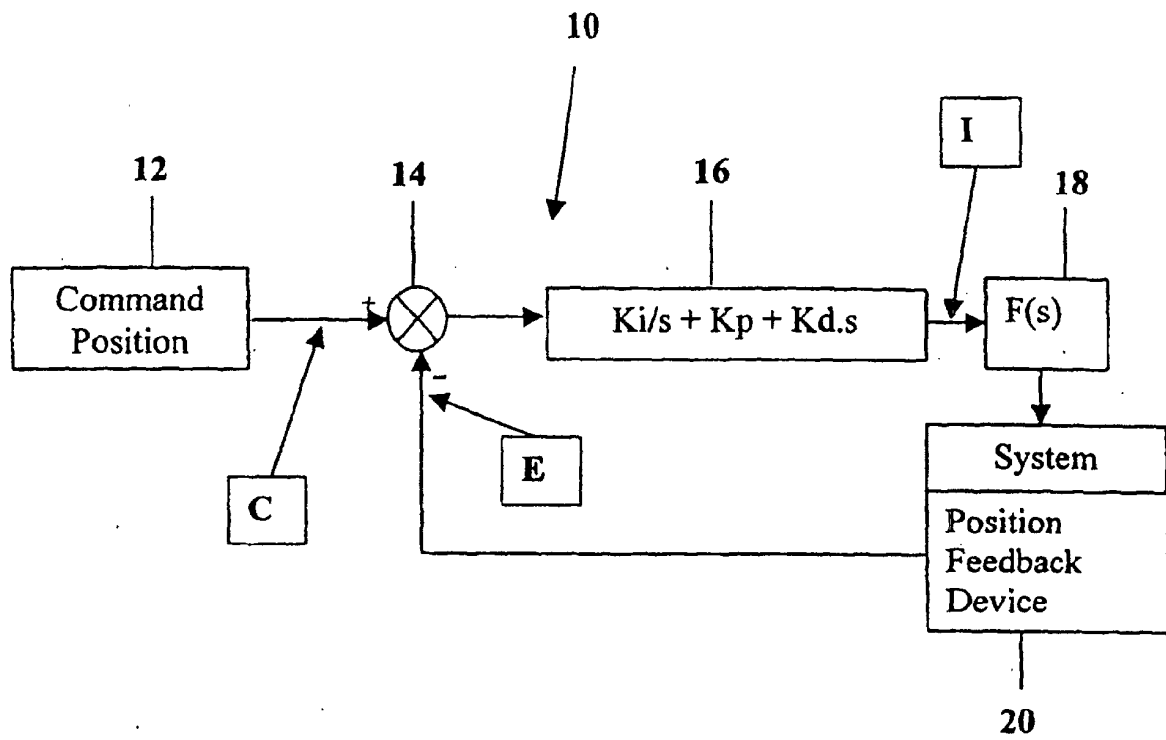


Fig 1

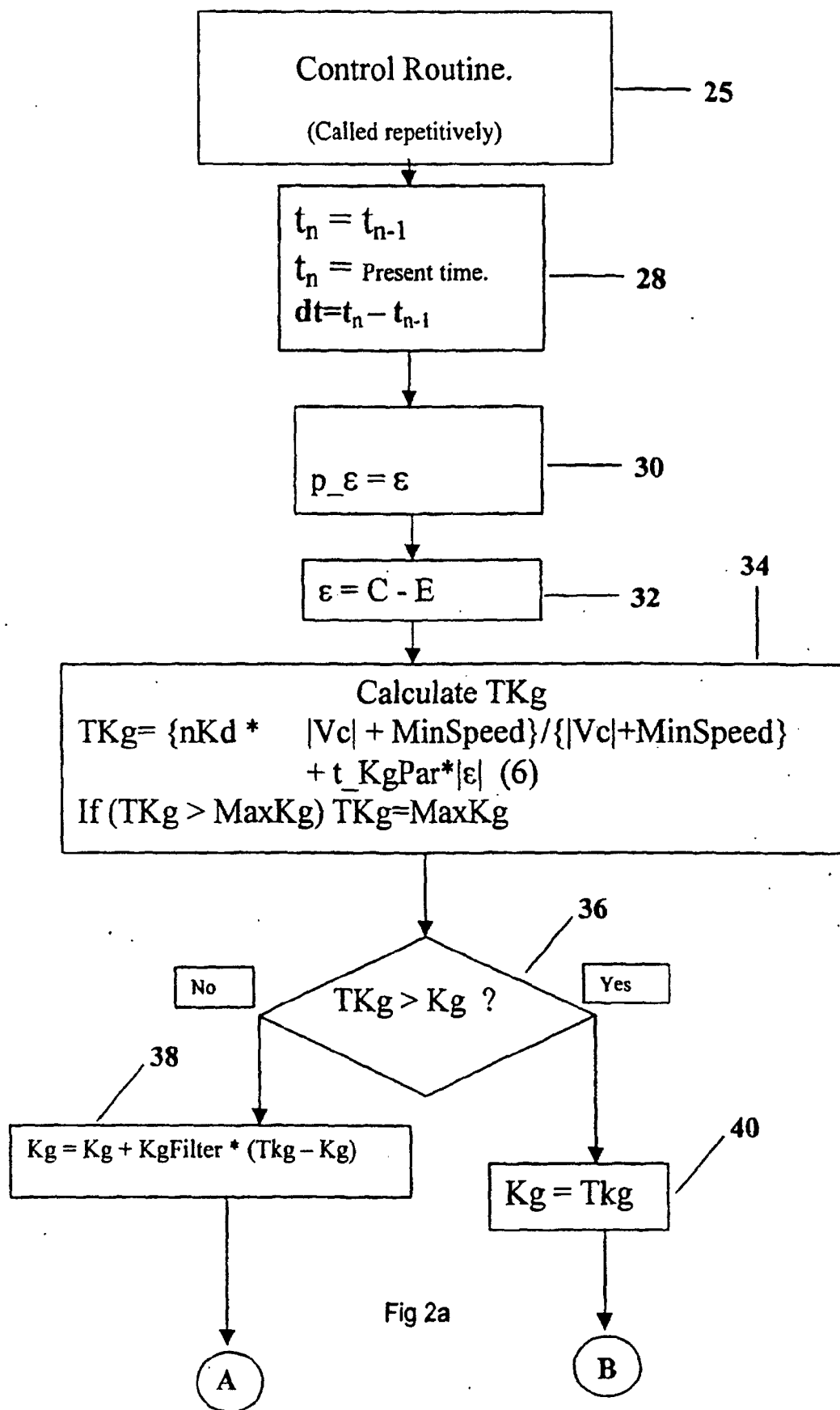


Fig 2a

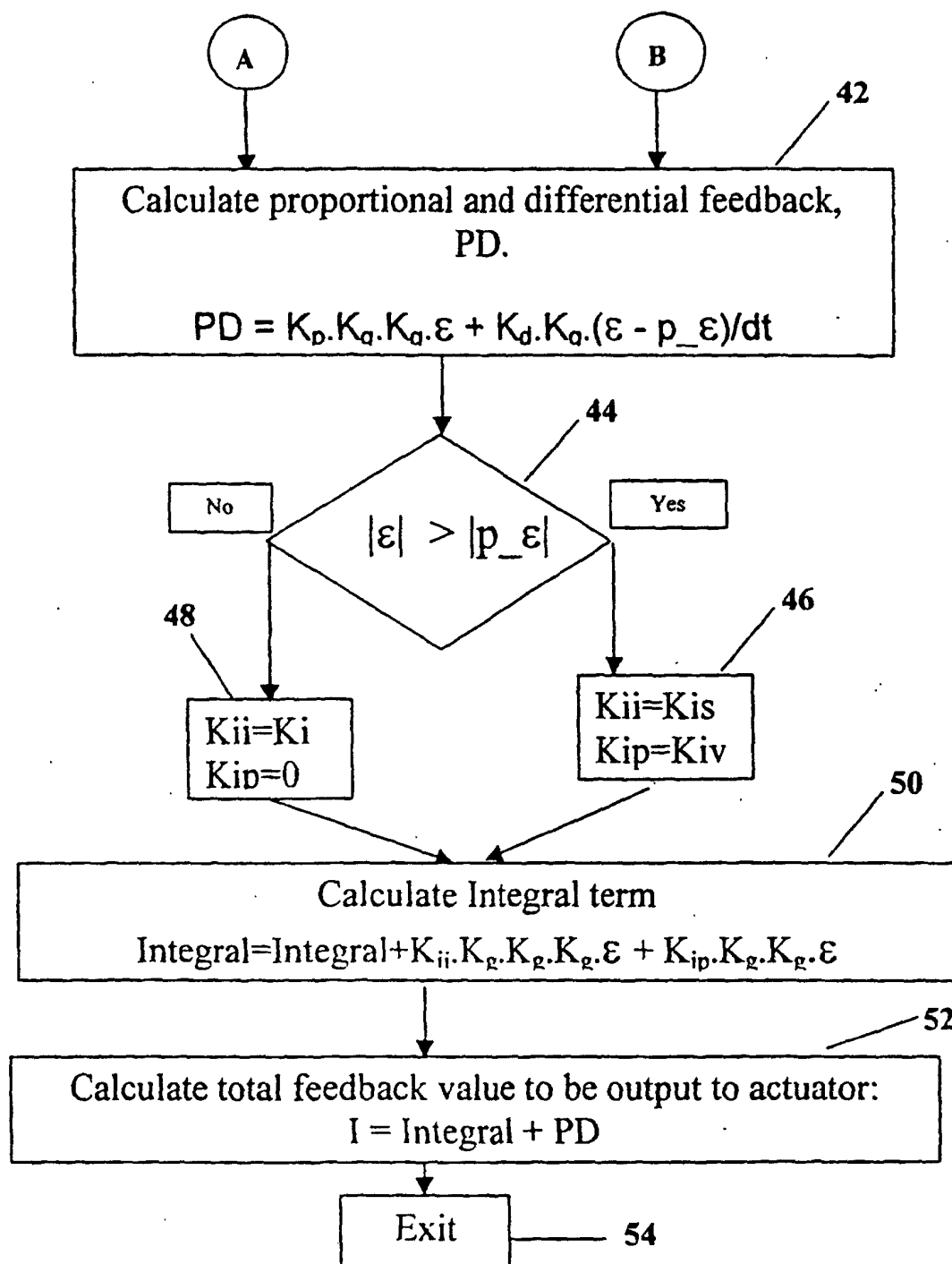


Fig 2b